

Week1 fundamentals

Cloud computing is the on-demand delivery of IT resources like computing power, storage, network, and databases over the internet. Instead of buying and maintaining physical data centers and servers, we rent access to these services from a cloud provider, paying only for what we use.

Types of cloud computing:

- Public: 3rd-party providers (Microsoft Azure, Amazon Web Services) deliver all the services(network,storage,db) over the internet.
- Private: Cloud computing resources are used by a single organisation. Located on company's on site datacenter.
- Hybrid: Public+Private. Allows data and apps to move safely between both the types.

Service Models:

- IaaS: we rent basic infrastructure (servers, network, storage) on a pay-as-you-go basis. We manage the OS and applications. Eg. AWS EC2, Google compute engine.
- PaaS: Gives a built-in environment to write n test code. Eg. Google App engine.
- SaaS: Delivers complete, working apps that we can use via a web browser. Eg. Gmail, Zoom.

Microsoft Azure:

Microsoft Azure is Microsoft's public cloud computing platform. It provides cloud services, computing power, storage, and networking, allowing organizations to build, manage and deploy applications.

Resource groups:

It is a logical container into which azure resources are deployed and managed.

Storage Accounts:

It contains all of the Azure storage data objects :blobs ,file shares, queues tables, and disks

Data Warehouse:

Central repository to store cleaned, structured and historical data for reporting and analysis.

Data Lake: It stores raw, unprocessed data in its native format.

Data Lake house combines both data warehouse and data lake

Hadoop:

An older, traditional method of handling big data. It splits huge task among dozen of small computers then writes everything on hard drive. Slow

Apache Spark:

Modern, replacement for Hadoop. It is super-fast engine to process huge amount of data. Open source,distributed. Huge task is divided into clusters of computers. Uses in-memory computation.

Spark Architecture:



Driver is the brain of the spark. It breaks the huge task into smaller tasks. Sends those tasks to workers.

Cluster manager allocates resources to workers.

Worker nodes executes the tasks through executors.

Databricks:

It is a cloud based data processing platform that reads huge amount of raw data, cleans it, transforms it and analyses it to generate useful insights. Built on Apache Spark. Collaborative notebooks (Python, SQL, Scala, R), Structured, semi-structured, and unstructured data support

Databricks Architecture

Databricks follows a Lakehouse Architecture, which combines the benefits of a Data Lake and a Data Warehouse.

2 layer architecture:

- control plane: manages web interfaces, notebooks, jobs.
- data plane : process the data

Workspace:

A **Workspace** is the environment where users collaborate in Databricks. It contains Notebooks, Dashboards, Libraries, Jobs.

Notebook:

A **Notebook** is an interactive development environment used to write and execute code.

Compute: Resources (CPU, Memory, Storage) used to execute notebooks, jobs, SQL queries, and ML tasks.

Delta Lake

Delta Lake is the storage layer of Databricks.

Features:

- ACID Transactions: atomicity, consistency, isolation, durability
- Data Versioning (Time Travel): Delta Lake keeps a new version instead of permanently replacing the old one. Suppose someone accidentally deletes data today. With Delta Lake, we can go back and view the data as it existed yesterday or last week.
- Schema Enforcement: DL checks whether the incoming data matches the expected schema.
- Schema Evolution: can change or add new columns without disturbing the existing columns.
- Scalable Metadata Handling: efficiently managing information about data, even when system contains billions of files.

Delta Lake Medallion Architecture

- Bronze Layer: Raw data ingestion.
- Silver Layer: Cleaned and validated data.
- Gold Layer: Business-ready aggregated data.

Azure Synapse: Microsoft's unified integrated **analytics** platform that brings together data storage, big data processing, SQL analytics, and reporting in a single workspace.

Data Warehouse + Data Lake Analytics + ETL + Big Data Processing in one platform

Azure Synapse	Azure Databricks
1.All-in-one analytics platform	1.Data engineering and data science platform .
2.Strong for SQL analytics and data warehousing.	2.Strong for Spark-based big data processing.
3.can query, store, and analyze data	3.cleans,transforms, and analyses data using spark.
4.Integrates with power bi	4.Integrates with ADLS, Synapse, and power bi

Git:

Version control system help to track changes in code. Free and open source, fast and scalable,tracks the history,collaborate.

Github:

Website that allows developers to store n manage their code using git.